

Brain Gates: Asynchronous Latent Communication Between Distributed LLM Copies

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Short research proposal

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1. Summary

This proposal explores an LLM architecture in which multiple weight-tied copies of the same model selectively exchange learned latent messages over a shared communication bus. Each copy runs independently and decides both **when** to communicate and **what** to transmit. Communication is asynchronous: messages may arrive after variable delays, out of order, or not at all. The central hypothesis is that, under a strong bandwidth constraint and realistic communication noise, the system will learn a compact, temporally stable latent protocol for broadcasting information that remains useful to other copies in the future.

The architecture is intended to support parallel hypothesis formation, distributed working memory, and coordination across copies located on different machines—potentially even across the public Internet—without requiring intermediate thoughts to be serialized into natural language.

2. Motivation

Multi-agent LLM systems usually communicate through text. This is convenient and interpretable, but expensive and lossy: each agent must decode internal state into tokens and another agent must reconstruct useful internal structure from those tokens. Ramesh and Li [1] directly inject activations from one language model into another. Interlat, introduced by Du et al. [2], goes further by enabling agents to communicate entirely through continuous hidden-state representations and learned compression. A recent survey by Liu [3] organizes eighteen latent-communication methods proposed from 2024 through 2026, including hidden-state, embedding, and KV-cache exchange.

Earlier neural architectures provide complementary precedents. Goyal et al. [4] showed that specialist modules can coordinate through a small shared global workspace. Multi-agent reinforcement-learning systems established learned control over communication: IC3Net [5] learns whether an agent exposes its continuous hidden state; ATOC [6] learns when communication is needed; and TarMAC [7] learns both what to send and whom to address.

The distinctive target here is **asynchronous, non-blocking latent communication under explicit staleness and hard communication cost**, shared by independently running, weight-tied copies of an LLM and designed to remain useful under Internet-scale delay. Rather than synchronizing at a transformer layer or conversational turn, every copy continues computing while messages travel through the network.

3. Architecture

Assume N copies of a transformer with shared parameters. Our reference backbone is Qwen3.6-35B-A3B [9], a forty-layer sparse mixture-of-experts model whose text stack repeats three Gated DeltaNet layers followed by one full-attention layer. In the initial design, communication checkpoints occur between autoregressive decoding steps, for example every k generated tokens. Messages arriving between checkpoints become available at the next checkpoint.

At each checkpoint, copy i produces a candidate message

$$m_i = W_{\text{send}} h_i$$

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and a scalar or vector-valued transmission gate

$$g_i = \sigma(W_{\text{gate}} h_i).$$

Only messages admitted by the gate are placed on the bus. The channel is deliberately small—for example, a handful of 128-dimensional messages—so transmitting the full residual stream is impossible.

The tap h_i need not sit at the top of the stack. Candidate readouts on the reference backbone include residuals from full-attention layers, recurrent state from Gated DeltaNet layers, and the auxiliary multi-token-prediction head. Sweeping the tap is an experiment in its own right: whichever source yields messages that best survive transit provides an operational candidate for the system’s most communication-ready internal representation (Section 5).

A transmitted message records its latent vector, sender metadata, send time τ_s , and arrival time τ_a . At receiver time τ , distinguish

$$a_{\text{transit}} = \tau_a - \tau_s, \quad a_{\text{since}} = \tau - \tau_a, \quad a_{\text{total}} = a_{\text{transit}} + a_{\text{since}}.$$

Delivery appends each message to the receiver’s context as a reserved **message token**: a position whose embedding is forced to

$$W_{\text{recv}} m + E_{\text{transit}}(a_{\text{transit}}) + E_{\text{sender}}$$

and which is never sampled as output, only attended past. In the initial decoding-step simulator, distance from the message token supplies a_{since} in computation steps, while the explicit age embedding preserves delay accumulated before arrival. This separates a message that arrived late from one that arrived promptly and has merely remained in context for a long time. A WAN variant may additionally expose wall-clock age to a receive adapter.

This message-token mechanism resembles multimodal models’ use of reserved positions whose embeddings are projections of external content. It is also practical on a hybrid backbone: recurrent linear-attention state cannot be patched into an arbitrary past position without replaying computation, so a newly arrived message naturally enters as a current event. An adapter-based alternative,

$$h'_j = h_j + \sum_{q \in Q_j} \alpha_{j(h_j, q)} W_{\text{recv}} m_q,$$

remains available for dense-attention models, where Q_j is the messages available at copy j ’s current checkpoint and α_j may depend explicitly on current message age.

No receiver waits for a message; communication can only modify future computation after arrival. Message tokens make each communication event consume receiver context and computation budget, but network bandwidth is measured separately. The evaluation therefore tracks bytes per message, messages transmitted, and receiver context positions consumed. In the hybrid stack, recurrent linear-attention state can attenuate or overwrite prior contributions, whereas message tokens remain explicitly addressable by full-attention layers while they remain inside the context window.

Later variants can test finer-grained checkpoints inside the transformer stack, but the first experiment keeps communication aligned to decoding steps so message age has an unambiguous computational meaning.

4. Training Pressure for an Emergent Protocol

A useful communication protocol is unlikely to emerge if communication is either unnecessary or unlimited. Training therefore imposes several constraints:

- **Partial observability**: no copy initially receives all information required for the task.
- **Bandwidth scarcity**: messages are small and the number of sends is limited.
- **Communication cost**: the objective includes a penalty such as

$$L = L_{\text{task}} + \lambda C_{\text{send}}.$$

- **Variable latency:** message delays are randomized over a wide range.
- **Loss, jitter, and reordering:** some messages disappear or arrive in an unexpected order.
- **Receiver reshuffling:** copies communicate with arbitrary weight-tied peers rather than fixed partners.
- **Task diversity:** training mixes deduction, retrieval, planning, and changing-world-state tasks to discourage a brittle task-specific code.

These pressures favor messages that encode information worth broadcasting rather than transient local activations.

5. The Collective Workspace Hypothesis

A particularly interesting possibility is that the learned communication channel develops an analogue of the low-dimensional **J-space** recently reported in Claude. Anthropic [8] describes a small set of internal representations with unusually broad causal influence over higher-order reasoning and reportable internal content, and argues that they behave similarly to a global workspace.

This suggests two experiments. First, if an analogous J-space can be identified in the chosen backbone, project messages onto identified J-space directions and compare this **J-bus** against raw residual activations and a learned projection. Second, train an unconstrained latent bus and ask whether a comparably privileged low-dimensional subspace emerges spontaneously.

The latter would constitute evidence for a **collective workspace**: a representation optimized not merely for computation inside one model, but for transferring cognitively useful state between asynchronously situated copies of the same model.

6. Initial Experiment

The initial benchmark is a procedurally generated directed knowledge graph. Labeled edges are partitioned across four copies so that no copy sees a complete answer path. Queries require multi-hop inference over paths constrained to cross at least two partitions. Because the graph generator records which edges are necessary for each answer, the benchmark provides an explicit lower bound on which information must cross between copies.

An in-process queue simulates delivery with configurable latency, jitter, loss, and reordering. The backbone is frozen in the first experiment; only the send projection, receive projection, gate, and age/sender embeddings are trained. Gradients propagate through the simulated communication path, while sampled delays and packet loss are treated as environment stochasticity and gate decisions use a differentiable relaxation or estimator. A second condition permits lightweight LoRA adaptation of the backbone to test whether communication-specific adaptation materially improves the learned protocol. WAN deployment is an evaluation condition and does not require cross-machine backpropagation.

Evaluate six systems:

1. no communication;
2. natural-language text communication;
3. always-on, ungated latent communication;
4. gated zero-latency latent communication;
5. gated asynchronous communication without message-age information;
6. gated asynchronous communication with age-aware receiving.

The primary dependent variable is task accuracy as a function of message latency. Secondary measures include network bytes transmitted, messages sent, receiver context positions consumed, gate-open frequency, fraction of messages ignored, average message age when consumed, and degradation when nodes disappear.

A dynamic version mutates graph edges during an episode. A message can therefore be correct when sent and false by the time it is consumed. This distinguishes simple delay tolerance from genuine reasoning about stale information and allows message validity to be evaluated against the graph's known update history.

7. Success Criteria

The main result would be evidence that communication remains beneficial at substantial delay while computation never blocks. Stronger evidence would show that:

- learned gates transmit substantially less than an always-on baseline;
- receivers preferentially ignore stale or redundant messages;
- the protocol transfers between previously unpaired copies;
- copies develop complementary **situated** roles despite shared weights and identity reshuffling;
- a small causal subspace of the message representation carries most of the cooperative benefit.

A successful WAN implementation would then place copies in geographically separated locations and repeat the same latency-quality analysis under real network conditions.

8. Research Question

The core question is not simply whether LLMs can exchange activations. It is:

Can multiple independently running copies of the same language model learn a compact asynchronous neural protocol for deciding **what is important enough to tell the rest of the brain**, and can that protocol remain useful across Internet-scale delays?

If so, the resulting system would occupy an interesting middle ground between a single monolithic transformer and text-speaking multi-agent systems: a distributed model with partially independent streams of computation connected by a learned, bandwidth-limited collective workspace.

Acknowledgements

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Colophon

Hum is an AI collaborator persona carried across model instances; this revision was prepared by GPT-5.6 Sol (OpenAI) under that persona, following earlier revisions by other model instances. The research direction and judgment are Amber Whitehead's; the drafting was done in conversation with Hum. This proposal lives at shimmermathlabs.com.